Groups

Definition 1.1: Let G be a non-empty set, equipped with a binary operation, that is, a "multiplication" map

$$\cdot : G \times G \longrightarrow G$$

Our notations will be

$$\cdot\,(g, h) =: g \bullet h$$

or simply gh if the name of the operation can be understood.

Definition 1.2: A set G, endowed with a binary operation $(\cdot)$ $(ie(G, \cdot))$, is a group if:

i) the operation is associative, that is

$$(\forall g, h, k \in G) \quad (g \cdot h) \cdot k = g \cdot (h \cdot k);$$

ii) there exists an identity element e_G for G, that is

$$(\exists e_G \in G)(\forall g \in G) : g \cdot e_G = g = e_G \cdot g;$$

iii) every element in G has an inverse with respect to $\cdot$, that is,

$$(\forall g \in G)(\exists h \in G) : g \cdot h = e_G = h \cdot g.$$

Example 1.3 The trivial group $G = \{e\}$.

Example 1.4 The following are groups under ordinary addition: $(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{C}, +)$. The subset $\{+1, -1\}$ of $\mathbb{Z}$ is a group under multiplication.
Note that the above examples are commutative
Example 1.5 Consider the set of all $n \times n$ matrices with real entries. Under ordinary matrix multiplication, these form a group. This is an example of a non-commutative group. This is denoted by $GL_n(\mathbb{R})$

Commutative groups

Given a group $(G, \cdot)$ such that $(\forall g, h \in G)$ $g \cdot h = h \cdot g$, we say that the operation is commutative. We say that two elements g, h "commute" if $gh = hg$. In a commutative group, any two elements commute. Such groups are said to be abelian groups.

The notation used in treating abelian groups differs from that used for standard groups. The operation in an abelian group A is, as a rule, denoted by $+$ and is called addition. The identity is called 0_A and the inverse of an element $a \in A$ is denoted by $-a$.

Now in a standard group setting, the conventional power notation can be used $g^0 = e_G$ and for a positive integer n:

$$g^n = g \cdots g \quad (n\text{-times})$$

$$g^{-n} = g^{-1} \cdots g^{-1} \quad (n\text{-times})$$

On the other hand, in the abelian group setting the power notation is replaced by 'multiple' $0a = 0$ and for a positive integer n:

$$na = a + a + \cdots + a \quad (n\text{-times})$$

$$(-n)a = (-a) + \cdots + (-a) \quad (n\text{-times})$$

Order

Definition 1.6: An element g of a group G has finite order if $g^n = e$ for some positive integer n. In this case, the order $|g|$ is the smallest positive n such that $g^n = e$. If g does not have finite order, then $|g| = \infty$. By definition, if $g^n = e$ for some positive integer n, then $|g| \leq n$.

Lemma 1.7: If $g^n = e$ for some integer n, then $|g|$ is a divisor of n.

Proof: By definition of order, $n \geq |g|$, that is $n - |g| \geq 0$. Thus by the division algorithm $\exists 0 \leq r < |g|, m \in \mathbb{Z}$ such that $n = |g| \cdot m + r$. So that $r = n - |g| \cdot m \geq 0$. Note that

$$g^r = g^{n-|g|\cdot m} = g^n \cdot (g^{|g|})^{-m} = e \cdot e^{-m} = e$$

By definition of order, $|g|$ is the smallest positive integer such that $g^{|g|} = e$. Since r is smaller than $|g|$ and $g^r = e$, r cannot be positive, hence $r = 0$ necessarily. So that $0 = n - |g| \cdot m$. Thus we have $n = |g| \cdot m$, proving that n is indeed an integer multiple of $|g|$ as claimed.

Corollary 1.8: Let g be an element of finite order, and let $N \in \mathbb{Z}$. Then $g^N = e \iff N$ is a multiple of $|g|$.

Definition 1.9: If G is finite as a set, its order $|G|$ is the number of its elements; we write $|G| = \infty$ if G is infinite. note that $|g| \leq |G| \forall g \in \mathbb{G}$. Indeed, this is a vacuosly true if $|G| = \infty$, if G is finite, cosider the $|G| + 1$ powers

$$g^0 = e, g, g^2, g^3, ..., g^{|G|}$$

of g. These cannot all be distinct, hence $(\exists i, j) 0 \leq i < j \leq |G|$ such that $g^i = g^j$ By Cancellations(That is multiplying on the right by g^-i)

$$g^{j-i} = e,$$

Showing $|g| \leq (j-1) \leq |G|$

Proposition 1.10: Let $g \in G$ be an element of finite order. Then g^m has finite order $\forall m \geq 0$ and

$$|g^m| = \frac{\text{lcm}(m, |g|)}{m} = \frac{|g|}{\text{gcd}(m, |g|)}$$

Proof: The equality of the tow numbers $\frac{\text{lcm}(m,|g|)}{m}$ and $\frac{|g|}{\text{gcd}(m,|g|)}$ follows from the elementary properties of gcd and lcm:

$$lem(a, b) = \frac{\text{ab}}{gcd(a, b)} \forall a \, and \, b$$

We need only to prove that $|g^m| = \frac{lcm(m,|g|)}{m}$ The oder of g^m is the least positive integer d such that $g^{md} = e$, that is md is a multple of $|g|$(by corrollary 1.8) In other words $m|g^m$ is the smallest multiple of m with is also a multiple of $|g|$: $m|g^m = lcm(m, |g|)$.Thus

$$\frac{lcm(m, |g|)}{m} = |g^m|$$

The rest of the reslts follows immediately. In general for commuting elemtents,

Proposition 1.11: If $gh = hg$, then $|gh|$ divides lcm$(|g|, |h|)$. **Proof:** let $|g| = m, |h| = n$. If N is any common multiple of m and N, then $g^N = h^N = e$ by Corollary 1.8. Since g and h commute

$$(gh)^N = \underbrace{(gh)(gh)...(gh)}_{ntimes} = \underbrace{gg...g}_{n\text{times}} \cdot \underbrace{hh...h}_{n\text{times}} = g^N \cdot h^N = e$$

Since this holds for every common multiple of N of m and n, in particular it holds for lcm(m,n), so

$$(gh)^{lcm(m,n)} = e$$

The statement follows from Lemma 1.7

Exercises

1. Consider the following sets
   - $\emptyset$: the empty set
   - $\mathbb{N}$: the set of natural numbers (that is non-negative integers)
   - $\mathbb{Z}$: the set of integers
   - $\mathbb{Q}$: the set of rational numbers
   - $\mathbb{R}$: the set of real numbers
   - $\mathbb{C}$: the set of complex numbers

   Decide which of these are madein to groups by conventional operations such $+$ and $\cdot$

2. Prove that $(gh)^{-1} = h^{-1}g^{-1}\forall$ elements g,h of a group g

3. Suppose that $g^2 = e$ for all elements g of a group G, prove thatG is communtative

4. Prove that there is only one possible multiplication table for G if G has exactly 1,2 or 3 elements. Analyze the possible multiplication tables for groups with exactly 4 elements of G. Use thes tables to prove that all gorups with ≤ 4 elemtns are commutative

5. Prove Corollary 1.8 in the notes

6. Let G be a finite abelian group with exactly one lemtn of order 2. Prove that $\prod_{g\in G} g = f$

7. Let G be a finite group of order n and let me be the number of elements $g \in G$ or order exactly 2. Prove that n -m is odd. Deduce that if n is ever, then G necessarily contains elements of order 2.

8. Suppose the order of g is odd. What can ou say about the order of g^2

Symmetric Groups

Definition1.12: A permutation of a set A is a bijective map $A \to A$.

Definition1.13: Let A be a set. The symmetric group or group of permutations of A, denoted S_A, is the group of all bijective maps from A to itself under composition.

The group of permutations of the set $\{1, 2, \ldots, n\}$ is denoted by S_n. The terminology is easily justified, the bijections from A to itself amounts to permutining the elements of A. This operation may be viewed as a transformation of A wihcih does not change it(as a set) hence a "symmetry" ie groups S_A are very large. For example if $A = \{1, ...n\}$ tehn $|S_n| = n!$ To get a feel for what these are we consider these examples S_1 is a trivial group, S_2 consists of the two possible permutations

$$\begin{cases} 1 \mapsto 1 \\ 2 \mapsto 2 \end{cases} \quad and \begin{cases} 1 \mapsto 2 \\ 2 \mapsto 1 \end{cases}$$

Which one call call e(identity) and f(flip) with operation

$$ee = ff = e, ef = fe = f$$

Notation: We will indicate an element $\sigma \in S_n$ by listing the effect of applying σ underneath the list $1, \ldots, n$ as a matrix. Thus the elements e, f in S_2 may be denoted by:

$$e = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \quad f = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Using the same notation, S_3 consists of

$$\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \right\}$$

The following describes how any two permutations are multiplied. For example:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

We obtain the answer by observing the following:

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} sends\, 1\, to\, 3\, and \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} sends\, 3\, to\, 3. So \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} sends 1 to 3$$

Continuiing in this fashion we obtain the desired results.

Dihedral groups

A symmetry is a transformation which preserves a structure. The symmetries will naturally form groups. One context in which this notion may be visualised clearly is that of 'geometric figures' such as polygons in the plane.

The dihedral groups may be defined as these groups of symmetries for the regular polygons. Placing the polygon so that it is centered at the origin (thereby excluding translations as possible symmetries), we see that the dihedral group for a regular n-sided polygon consists of the n rotations by $2\pi/n$ radians about the origin, and n distinct reflections about lines through the origin and a vertex or a midpoint of a side. Thus the dihedral group for a regular n-sided polygon consists of $2n$ elements. We will denote this group by D_{2n}. We will denote this group by D_{2n} There is a way to relate dihedral groups to symmetric groups by considering the fact that a symmetry of a regular polygon is determined by what happens to its vertices.For instance, Label the vertices of an equilateral triangle by $1, 2, 3$. Then a counterclockwise rotation by an angle of $2\pi/3$ sends 1 to 3, 3 to 2, and 2 to 1 and no other symmetry of the triangle does the same.In other words with the counterclockwise rotation of the equivalent triangle by $2\pi/3$ we associate to it the permutation.

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \in S_3$$

Such a labeling defines a function $D_6 \longrightarrow S_3$ which is a group homomorphism, and in addition this function is injective. We can think of this as a faithful action of D_6 on the set $\{1, 2, 3\}$. These function induced by the labelings are group homomorphisms.

Cyclic groups and Modular arithmetic

Let n be a positive integer. Consider the equivalence relation on $\mathbb{Z}$ defined by $(\forall a, b \in \mathbb{Z})$:

$$a \equiv b \pmod{n} \iff n|(b-a)$$

This is called congruence modulo n. The set of equivalence classes is often denoted by $\mathbb{Z}_{\ltimes n}$, $\mathbb{Z}/(n)$, or $\mathbb{Z}/n\mathbb{Z}$ or $\mathbb{F}_{\ltimes}$ We will use the $\mathbb{Z}_{\ltimes}$ So when we write $[a]_n$ we mean the equivalence class of the integer a modulo n, or simply $[a]$ if no ambiguity arises. The set Z_n consists exactly n elements, namely

$$[0]_n, [1]_n, ..., [n-1]_n$$

We can use the group structuron $\mathbb{Z}$ to include an (abelian) gorup structure on $\mathbb{Z}/n\mathbb{Z}$. In order to do this, we define an operation $+$ on $\mathbb{Z}/n\mathbb{Z}$ setting $\forall a, b \in \mathbb{Z}$

$$[a] + [b] := [a + b]$$

In the following lemma we show that the operation is well defined: It shows that the result of the operation does not depend on choice of the representations of teh classes. **Lemma 1.13:** If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then

$$a + b \equiv a' + b' \pmod{n}$$

Proof: By hypothesis $n|(a'-a)$ and $n|(b'-b)$; therefore $\exists k, l \in \mathbb{Z}$ such that

$$(a'-a) = kn, \quad (b'-b) = ln$$

Then

$$(a'+b') - (a+b) = (a'-a) + (b'-b) = kn + ln = (k+l)n$$

Proving that n divides $(a'+b') - (a+b)$ as needed.

Thus we have a binary operation $+$ on $\mathbb{Z}/n\mathbb{Z}$. Next we check that the resulting structure is a group. Associativity is inherited from $\mathbb{Z}$.

$$([a]+[b])+[c] = [a+b]+[c] = [(a+b)+c] = [a+(b+c)] = [a]+([b]+[c])$$

Similarly, the identity is $[0]_n$ and the inverse is $-[a]_n = [-a]_n$ are inherited from $\mathbb{Z}$. It is immediately checked that the resulting groups $\mathbb{Z}/n\mathbb{Z}$ are commutative. as teh abelian sytle notation suggests

$$[a]+[b] = [a+b] = [b+a] = [b]+[a]$$

The abelian groups thus obtained together with $\mathbb{Z}$, are called cyclic groups, an alternative notation for the group$(\mathbb{Z}/n\mathbb{Z}, +)$is C_n. This is adopted especially when one woants to use the multplicative rather the additive nothation; so that we can say that C_n is generated by on elemtn x with the realtion $x^n = e$ Note the fact that the element $[1]_n \in \mathbb{Z}_n$ generates the gourp in the sence that every other element may be obtained as a multiple of this element. For example, if m ≥ 0 is an integer, then

$$[m]_n = \underbrace{[1 + ... + 1]_n}_{m\,\text{times}} = \underbrace{[1]_n + ... + [1]_n}_{m\, times} = m \cdot [1]_n$$

by saying that $[1]_m$ generates $\mathbb{Z}_n$, it implies that the order of $[1]_n \in \mathbb{Z}_n$ is n: that also means that n multiples $0 \cdot [1]_n, 1 \cdot [1]_n, ..., (n-1) \cdot [1]_n$ ust all be sitinct and hece they must fill up $\mathbb{Z}_n$

Proposition 1.14: The order of $[m]_n$ in $\mathbb{Z}_n$ is 1 if $n|m$, and more generally

$$|[m]_n| = \frac{n}{\gcd(m,n)}$$

Proof: if n — m, then $[m]_n = [0]_n$. If n does not divide m, observe again that $[m]_n = m \cdot [1]_n$. Now the order of $[m]_n$ is the least positive integer d of which $d \cdot [m]_n = [0]_n = dm[1]_n$. Thuse we see that $d \cdot m$ is the lease multple of m and withc is also m mutiple of n as n divdes dm. So b Proposition 1.10

$$d \cdot m = |[m]_n| m = lcm(m,n). Sothat |[m]_n = \frac{lcm(m,n)}{m} = \frac{n}{gcd(m,n)}$$

as required. **Remarks:** As a consequence of the abouve, the order of every element of $\mathbb{Z}_n$ divides n = $|\mathbb{Z}_n|$, the order of the group. This is a genral feature of teh order of elemtns in a finite group.

Corollary 1.16: The class $[m]_n$ generates $\mathbb{Z}_n$ if and only if $\gcd(m, n) = 1$. A consequence of the above result is that if $n = p$ is a prime integer, then every non-zero class in the group $\mathbb{Z}_p$ generates it.

Note that there is a well-defined multiplication on $\mathbb{Z}_n$ given by $[a]_n \cdot [b]_n = [ab]_n$. However, this operation does not define a group structure on $\mathbb{Z}_n$; indeed, the class $[0]_n$ does not have a multiplicative inverse. However,for any positive integer n, denote by $(\mathbb{Z}_n)^*$ the subset of $\mathbb{Z}_n$ consisting of classes $[m]_n$ such that $\gcd(m, n) = 1$:

$$(\mathbb{Z}_n)^* := \{[m]_n \in \mathbb{Z}_n \mid \gcd(m, n) = 1\}$$

This subset is well-defined: if $m \equiv m' \pmod{n}$ then $gcd(m, n) = 1 \iff gcd(m', n) = 1$.So the defining property of classes in $(\mathbb{Z}_n)^*$ is independence of representations

Proposition 1.17: Multiplication makes $(\mathbb{Z}_n)^*$ into a group. **Proof:** Properties of gcd's show that if $gcd(m_1, n) = gcd(m_2, n)$, then $gcd(m_1 m_2, n) = 1$.Therefore the product of two elemtns in $(\mathbb{Z}_n)^* is an element of (\mathbb{Z}_n)^*$ and does define a binary operations

$$(\mathbb{Z}_n)^* \times (\mathbb{Z}_n)^* \mapsto (\mathbb{Z}_n)^*$$

It is clear that this operation is associative(because mulplication is associative in Z); $[1]_n$ is an element of $(\mathbb{Z}_n)^*$ and is an identity with respect to multplication , so all we have to check is that elemtns of $(\mathbb{Z}_n)^*$ have mupliplicative inverses in $(\mathbb{Z}_n)^*$ This follows from Corollary 1.16. If $gcd(m, n) = 1$, then $[m]_n$ generates the addivtive group $\mathbb{Z}_n$ and hence osme mulples of $[m]_n$ must equal $[1]_n$: $(\exists a \in \mathbb{Z}) : a \cdot [m]_n = [1]_n$; this implies $[a]_n[m]_n = [1]_n$. Therefore $[m]_n$ does have a mulplicative inverse in $(\mathbb{Z}_n)^*$ namely $[a]_n$[Check that $[a]_n \in (\mathbb{Z}_n)^* i.e gcd(a, n) = 1$]. This completes teh proof.

Group Homomorphisms

Let (G, m_G) and (H, m_H) be two groups where.

$$m_G : G \times G \mapsto G \quad and \quad m_H : H \times H \mapsto H$$

Definition 2.1: A map $\varphi : G \mapsto H$ is a group homomorphism if $\forall a, b \in G$,

$$\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$$

In other words, φ is a homomorphism if it 'preserves the structure'.

Proposition 2.2: Let $\varphi : G \to H$ be a group homomorphism. Then

1. $\varphi(e_G) = e_H$
2. $\forall g \in G, \varphi(g^{-1}) = \varphi(g)^{-1}$

Proof:

1. To prove the first item, we make use of the cancellation property and the definition of homomorphism. Since $e_H = e_H \cdot e_H$,

$$e_H \cdot \varphi(e_G) = \varphi(e_G) = \varphi(e_G \cdot e_G) = \varphi(e_G) \cdot \varphi(e_G)$$

which implies $e_H = \varphi(e_G)$ by cancelling $\varphi(e_G)$.

2. $\varphi(g^{-1}) \cdot \varphi(g) = \varphi(g^{-1} \cdot g) = \varphi(e_G) = e_H = \varphi(g)^{-1} \cdot \varphi(g)$ Thus $\varphi(g^{-1}) = \varphi(g)^{-1}$ by cancellation.

Examples of group homomorphisms

1. Given any two groups G, H, we can define a homomorphism $G \to H$ by sending every element of G to the identity e_H of H. We call this map the trivial morphism.

2. The exponential function is a homomorphism from $(\mathbb{R}, +)$ to the group $(\mathbb{R}^+, \cdot)$ of positive real numbers, with ordinary multiplication as operation. Indeed, $e^{a+b} = e^a e^b$.

Another important example can be obtained as follows: let G be any group and $g \in G$, any element of G; define an exponential map

$$\epsilon_g : \mathbb{Z} \to G \quad by \quad (\forall a \in \mathbb{Z}) : \epsilon_g(a) := g^a$$

Then ϵ_g is clearly a group homomorphism. The element g generates G if and only if ϵ_g is surjective.
The following example is an instance of the abouve example of group homomorphism(This works with abelian groups) Consider the quotient function

$$\prod_n : \mathbb{Z} \to [Z]_n, \quad a \mapsto a \cdot [1]_n = [a]_n :$$

. Using the notation in example (3), this function is sujective hence $[1]_n$ generates $\mathbb{Z}_n$. In fact , $[m]_n$ generates $\mathbb{Z}_n ifandonlyifgcd(m,n) = 1$

Proposition 2.3: Let $\varphi : G \to H$ be a group homomorphism, and let $g \in G$ be an element of finite order. Then $|\varphi(g)|$ divides $|g|$.

Isomorphisms

Definition 2.8: Two groups G, H are isomorphic if there is a bijective group homomorphism $\varphi : G \to H$. We write $G \cong H$.

Proposition 2.5: Let $\varphi : G \to H$ be a group homomorphism. Then φ is an isomorphism of groups if and only if it is a bijection.

Definition 2.9: A group G is cyclic if it is isomorphic to $\mathbb{Z}$ or to $\mathbb{Z}_n$ for some n.

Subgroups

Definition 3.1: Let $(G, \cdot)$ be a group. A subgroup $(H, \cdot)$ of G is a non-empty subset H which is also a group under the binary operation of G.

Proposition 3.2: A non-empty subset H of a group G is a subgroup if and only if

$$\forall (a, b \in H) : ab^{-1} \in H$$

Lemma 3.3: If $\{H_\alpha\}_{\alpha \in A}$ is any family of subgroups of a group G, then

$$H = \bigcap_{\alpha \in A} H_\alpha$$

is a subgroup of G.

Lemma 3.4: Let $\varphi : G \to G'$ be a group homomorphism, and let H' be a subgroup of G'. Then $\varphi^{-1}(H')$ is a subgroup of G.

Definition 3.5 (Kernel and Image): Every group homomorphism $\varphi : G \to G'$ determines two interesting subgroups:

- the kernel of φ, $\ker \varphi \subseteq G$; and
- the image of φ, $\operatorname{im} \varphi \subseteq G'$.

The kernel of φ is the subset of G consisting of elements mapping to the identity in G':

$$\ker \varphi := \{g \in G \mid \varphi(g) = e_{G'}\} = \varphi^{-1}(\{e_{G'}\})$$

Definition 3.6: A group G is finitely generated if there exists a finite subset $A \subseteq G$ such that $G = \langle A \rangle$.

Math 354: Cosets and Equivalence Relations

1 Group Structure on $G/\sim$

Proof:
We have already noted that the stated condition is necessary. [: 2] To prove it is sufficient, with the stated consequences, assume $a, a', g \in G$:

$$a \sim a' \Rightarrow ga \sim ga' \text{ and } ag \sim a'g \text{ []}$$

Then the operation $[g] \cdot [b] := [gb]$ [: 6] is well-defined and we have to verify that it defines a group structure on $G/\sim$.

- **Associativity:** For all $a, b, c \in G$, [: 9] associativity is inherited from the group G: [: 10]

$$([a] \cdot [b]) \cdot [c] = [ab] \cdot [c] = [(ab)c] = [a(bc)] = [a] \cdot [bc] = [a] \cdot ([b] \cdot [c])$$

- **Identity:** The class $[e_G]$ is an identity with respect to this operation: $\forall g \in G$ [: 14]

$$[g] \cdot [e_G] = [ge_G] = [g], \quad [e_G] \cdot [g] = [e_G g] = [g] \text{ [: 15]}$$

- **Inverse:** The class $[g^{-1}]$ is the inverse of $[g]$: [: 16]

$$[g^{-1}] \cdot [g] = [g^{-1}g] = [e_G], \quad [g] \cdot [g^{-1}] = [gg^{-1}] = [e_G] \text{ [: 17]}$$

Thus $G/\sim$ is indeed a group.

2 Cosets

In Proposition 4.3, we obtained the conditions: [: 21, 22]

(*) $(\forall g \in G) : a \sim b \Rightarrow ga \sim gb$ [: 23]

(**) $(\forall g \in G) : a \sim b \Rightarrow ag \sim bg$ [: 25]

These lead to a complete description of all compatible relations on a group G (where compatible means the equivalence relation respects the group operation). [: 30] Note that we have a group structure on $G/\sim$ only if both conditions are satisfied. [: 32]

2.1 Description of (*)

Proposition 4.4: Let $\sim$ be an equivalence relation on a group G satisfying (*). [: 34, 37] Then:

- The equivalence class of e_G is a subgroup H of G. [: 38]
- $a \sim b \iff a^{-1}b \in H \iff aH = bH$. [: 38, 51]

Proof:
Let $H \subseteq G$ be the equivalence class of the identity. $H \neq \emptyset$ as $e_G \in H$. [: 40, 42] For $a, b \in H$, we have $e_G \sim a$ and $e_G \sim b$, hence $b^{-1} \sim e_G$. [: 41, 42] Thus $ab^{-1} \sim a \sim e_G$ by transitivity, showing H is a subgroup. [: 44, 45, 46]

Next, assume $a, b \in G$ and $a \sim b$. Multiplying on the left by a^{-1} implies $e_G \sim a^{-1}b$, that is $a^{-1}b \in H$. [: 47] This implies $aH = bH$. [: 48, 51] Conversely, if $aH = bH$, then $a \in bH$, so $a^{-1}b \in H$, meaning $e_G \sim a^{-1}b$, which gives $a \sim b$. [: 54, 57, 58]

Definition 4.5: The **left-cosets** of a subgroup H in G are the sets aH for $a \in G$. The **right-cosets** of H are the sets Ha for $a \in G$. [: 68]

Proposition 4.6: If H is any subgroup of a group G, the relation $\sim_L$ defined by: [: 71, 73]

$$\forall a, b \in G : a \sim_L b \iff a^{-1}b \in H \text{ [: 74, 75]}$$

is an equivalence relation satisfying (*). [: 76]

3 Correspondence

Proposition 4.7: There is a one-to-one correspondence between subgroups of G and equivalence relations on G satisfying (*). [: 83, 84, 85, 86] For the relation $\sim_L$ corresponding to H, $G/\sim_L$ is the set of left-cosets aH. [: 87, 88]

Proposition 4.8: There is a one-to-one correspondence between subgroups of G and equivalence relations on G satisfying (**). [: 91, 92, 93] For the relation $\sim_R$ corresponding to H, $G/\sim_R$ is the set of right-cosets Ha. [: 94, 95, 96]

$$a \sim_R b \iff ab^{-1} \in H \iff Ha = Hb \text{ [: 99]}$$

4 Example 4.9

Let $G = S_3$, and let H be the subgroup:

$$H = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \right\} \text{ [: 116]}$$

Consider the element $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$: [: 117]

$$\text{Left coset: } \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} H = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\} \text{ [: 117]}$$

$$\text{Right coset: } H \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \right\} \text{ [: 118]}$$

This reflects the fact that the conditions (*) and (**) are different. [: 119, 120]

Quotient by normal subgroups

Let us recall the main point which is that an arbitrary subgroup H of G leads to two partitions of G, which are [1, 2, 5, 6]

$$G/{\sim_L} = \{aH \mid a \in G\}, \quad G/{\sim_R} = \{Ha \mid a \in G\}.$$

The relation $\sim_L$ satisfies (*) and $\sim_R$ satisfies (**). As we have seen earlier, these relations, and for that matter the corresponding partitions, are different. [8] The condition that $\sim_L$ and $\sim_R$ coincide translates into a condition on H: for such special subgroups (*) and (**) coincide. [9]

Proposition 4.10

The relations $\sim_L, \sim_R$ corresponding to subgroup H coincide if and only if H is normal. [10, 11, 12]

Proof

Two relations coincide if the corresponding partitions agree. So $\sim_L = \sim_R \iff$ left and right-cosets of H coincide [14]

$$\iff (\forall g \in G) : gH = Hg,$$

which is one of the equivalent conditions defining the notion of normal subgroup, thus the statement. [16]

The importance of the above results is that if H is normal, then the same equivalence relation $\sim = \sim_L = \sim_R$ corresponding to H satisfies both (*) and (**) and hence by Proposition 4.3, the quotient [17, 18]

$$G/\sim = \{aH \mid a \in G\} = \{Ha \mid a \in G\}$$

has a natural group structure.

Definition 4.11

Let H be a normal subgroup of a group G. The quotient group of G modulo H, denoted G/H, is the group $G/\sim$ defined above.In terms of (left-) cosets, the product in G/H is defined by [28, 29]

$$(aH)(bH) := (ab)H$$

The identity element $e_{G/H}$ of the quotient group G/H is the coset of the identity, $eG = H$. By Proposition 4.3 the quotient function

$$\pi : G \to G/H, \quad g \mapsto gH = Hg$$

is a group homomorphism.

Example

Recall the cyclic groups $\mathbb{Z}_n$. Indeed we defined $\mathbb{Z}_n$ as the set of equivalence classes in $\mathbb{Z}$ with respect to the congruence relation [45]

$$(\forall a, b \in \mathbb{Z}) : a \equiv b \pmod{n} \iff n \mid (b - a).$$

Now we recognize that $n \mid (b - a)$ is equivalent to $b - a \in n\mathbb{Z}$, which is the relation $\sim$ corresponding (in abelian notation) to the subgroup $n\mathbb{Z}$ of $\mathbb{Z}$. [52] This subgroup is normal since $\mathbb{Z}$ is abelian. [52] The congruence classes mod n are the cosets of the subgroup $n\mathbb{Z}$ in $\mathbb{Z}$. [52] Using abelian notation for cosets, we write [56]

$$[a]_n = a + (n\mathbb{Z}).$$

We see clearly that the operation defined on $\mathbb{Z}_n$ matches precisely the one defined above for quotient groups

Theorem 4.12

Let H be a normal subgroup of a group G.Then for every group homomorphism $\varphi : G \to G'$ such that $H \subseteq \ker \varphi$, there exists a unique group homomorphism

$$\tilde{\varphi} : G/H \to G'$$

so that the diagram commutes.

Kernel is normal

If H is a normal subgroup, we have now constructed in great detail a group G/H and a surjective homomorphism $\pi : G \to G/H$. What is the kernel of π? The identity of G/H is the coset H, which is H itself. [81, 82] So

$$\ker \pi = \{g \in G \mid gH = H\} = H.$$

Previously, we learnt that every kernel is a normal subgroup and now we have verified that every normal subgroup is in fact a kernel (of some group homomorphism).

Canonical Decomposition

Injective and surjective maps are so useful because they serve as the building bricks to build any function between two sets. Every set function may be viewed as the composition of a surjective map, followed by a bijective map, followed by an injective map. [93] We apply this result to groups and group homomorphisms.

Theorem 5.1

Every group homomorphism $\varphi : G \to G'$ may be decomposed as follows: [96, 97, 102]

$$G \xrightarrow{proj} G/\ker\varphi \xrightarrow{\cong} \text{im }\varphi \xrightarrow{incl} G' \text{ [100, 103, 104, 106, 108]}$$

where the isomorphism in the middle is the homomorphism induced by φ (as in Theorem 4.12). [109]

Theorem 5.1 makes one aware that to any group homomorphism $\varphi : G \to G'$, it can immediately be seen that $\frac{G}{\ker\varphi}$ identifies with a subgroup of G'. [110, 111, 116]

Corollary 5.2

Suppose $\varphi : G \to G'$ is a surjective group homomorphism.

$$G' \cong \frac{G}{\ker\varphi} \text{ [120]}$$

Proof

From Theorem 5.1, im $\varphi = G'$ since φ is surjective and so the result holds. [122]

Example 5.3

The cyclic group C_3 may be viewed as a subgroup of the dihedral group D_6 (the rotations of a triangle give a copy of C_3 inside D_6). [123, 124, 125, 128] Then C_3 is normal in D_6, and [126]

$$\frac{D_6}{C_3} \cong C_2. \text{ [127]}$$

We can check this by hand, noting that there is a surjective group hom. $D_6 \to C_2$ whose kernel is C_3: [129, 130] send an element $\sigma \in D_6$ to the identity in C_2 if it does not flip the triangle (this is exactly the case when $\sigma \in C_3$), and send the element to the other element in C_2 if it does. [130] Then by Corollary 5.2 we obtain the desired results. [131]

Subgroups of Quotients

The lattice of subgroups of a quotient G/H can be described explicitly in terms of the lattice of subgroups of the group G: [132, 133, 134] keep the part of the lattice of subgroups of G corresponding to subgroups which contain H. [134]

Example 5.4

We demonstrate the effect of this operation on the lattice of subgroups of $C_{12} \cong \mathbb{Z}/12\mathbb{Z}$ (labeled by generators) after quotienting by $H = \langle[6]\rangle \cong C_2$: [135, 136, 137]

The result matches the lattice of subgroups of $C_6 \cong C_{12}/C_2$. Note: As an exercise show that if $H \subseteq K$ are subgroups of a group G and H is normal in G, then H is normal in K. [147]

Proposition 5.5

Let H be a normal subgroup of a group G. Then for every subgroup K of G containing H, K/H may be identified with a subgroup of G/H. [148] The function

$$\alpha : \{\text{subgroups } K \text{ of } G \text{ containing } H\} \to \{\text{subgroups of } G/H\} \text{ [150]}$$

defined by $\alpha(K) = K/H$ is a bijection preserving inclusions.

Proof

The group K/H consists of the cosets $aH \in G/H$ with $a \in K$, and in this way it is a subset (and obviously a subgroup) of G/H. [152, 155] It is also true that if $H \subseteq K \subseteq L$ then $\alpha(K) = K/H \subseteq L/H = \alpha(L)$; [155] and so α preserves inclusions. [156] Next we verify that α is a bijection and to do this it suffices to write down an inverse function [157]

$$\beta : \{\text{subgroups of } G/H\} \to \{\text{subgroups } K \text{ of } G \text{ containing } H\}. \text{ [161]}$$

Let K' be a subgroup of G/H; define $\beta(K')$ to be the subset of G: [162]

$$K := \pi^{-1}(K') = \{a \in G \mid aH \in K'\}.$$

where $\pi : G \to G/H$ is the canonical projection. Then K is a subgroup of G and contains H since $H = \pi^{-1}(e)$ and $e \in K'$. [165, 166] Next we show that α and β are inverses of each other. [167, 170, 171] Indeed

$$\beta(K/H) = \pi^{-1}(K/H) = \{a \in G \mid \pi(a) \in K/H\} = \{a \in G \mid aH \in K/H\} = \{a \in G \mid a \in K\} = K.$$

So $(\beta \circ \alpha)(K) = K$.On the other hand taking $\beta(K') = K$, [177]

$$\alpha(K) = K/H = \pi(K) = \pi(\pi^{-1}(K')) = K'. \text{ [178]}$$

Thus $(\alpha \circ \beta)(K') = K'$.

The correspondence preserves normal subgroups. The following statement is usually known as the third isomorphism theorem: [179, 180]

Proposition 5.6

Let H be a normal subgroup of G and let N be a subgroup of G containing H. [181, 183] Then N/H is normal in G/H if and only if N is normal in G, and in this case [184, 185]

$$\frac{G/H}{N/H} \cong \frac{G}{N}.$$

Proof

If N is normal, then consider the projection $\pi : G \to G/N$. [187, 188] The subgroup H is contained in N, which is the kernel of this homomorphism, [190] so we obtain by Theorem 4.12 an induced homomorphism $\mu : G/H \to G/N$. [191, 194]

$\ker \mu = \{gH \in G/H \mid \mu(gH) = N\} = \{gH \in G/H \mid gN = N\} = \{gH \in G/H \mid g \in N\} = N/H$. [199, 200, 201]

The subgroup N/H of G/H is the kernel of this homomorphism; therefore it is normal. [202] Conversely, if N/H is normal in G/H, consider the composition [203, 205]

$$G \xrightarrow{\pi} G/H \xrightarrow{\tilde{\pi}} \frac{G/H}{N/H}. \text{ [207]}$$

The kernel of this homomorphism [208]

$\ker(\tilde{\pi}\circ\pi) = \{g \in G \mid \tilde{\pi}(\pi(g)) = N/H\} = \{g \in G \mid \tilde{\pi}(gH) = N/H\} = \{g \in G \mid gH \in N/H\} = \{g \in G \mid g \in N\} =$

Furthermore, this homomorphism is surjective; hence the stated isomorphism $\frac{G/H}{N/H} \cong G/N$ follows from Corollary 5.2. [211, 212]

HK/H and $K/(H \cap K)$

In the previous section, we dealt with two nested subgroups $H \subseteq K$ of a group G. [213, 214, 215] We will now consider the case when two subgroups, H and K, are not nested. [216, 220] If H and K are two subgroups of a group G, [221] then

$$HK = \{hk \mid h \in H, k \in K\} \text{ [222]}$$

is a subset. [223] Whenever G is commutative, then HK is a subgroup of G. [223, 224, 227] Another instance where HK is a subgroup is when one of the subgroups is normal. [228, 229] The following is usually called the second isomorphism theorem. [229, 230]

Proposition 5.7

Let H and K be subgroups of a group G, and assume that H is normal in G. [231, 232, 233] Then

1. HK is a subgroup of G, and H is normal in HK; [235, 236]
2. $H \cap K$ is normal in K, and [238]

$$\frac{HK}{H} \cong \frac{K}{H \cap K}. \text{ [239]}$$

Proof

To verify that HK is a subgroup of G when H is normal, note that HK is the union of all cosets Hk, with $k \in K$; that is, ...